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Stable thermophoretic trapping of generic particles at low pressures

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We demonstrate levitation and three-dimensionally stable trapping of a wide variety of particles in a vacuum through thermophoretic force in the presence of a strong temperature gradient. Typical sizes of the trapped particles are between $10\text{ }\mu\text{m}$ and 1 mm at a pressure between 1 and 10 Torr. The trapping stability is provided radially by the increasing temperature field and vertically by the transition from the free molecule to hydrodynamic behavior of thermophoresis as the particles ascend. To determine the levitation force and test various theoretical models, we examine the levitation heights of spherical polyethylene spheres under various conditions. A good agreement with two theoretical models is concluded. Our system offers a platform to discover various thermophoretic phenomena and to simulate dynamics of interacting many-body systems in a microgravity environment. *Published by AIP Publishing.* [<http://dx.doi.org/10.1063/1.4974489>]

Levitation of particles in ground-based experiments provides an ideal platform for the study of their dynamics and interactions in a pristine isolated environment and has broad applications in fundamental and applied sciences. Electromagnetically levitated atoms, ions, and molecules have opened the field of quantum degenerate gases,^{1,2} cold chemistry,³ and atom-based quantum information.⁴ Optical levitation of micron-size objects⁵ provides a new playground to investigate aerosols in atmosphere⁶ and Brownian motion of microspheres.⁷

Levitation of macroscopic particles in low pressure is of particular interest due to its wide applications in space, atmospheric, and even astro-chemical research. Several schemes to levitate glass spheres and water droplets have been realized based on optical tweezers,⁸ high magnetic field gradients,^{9,10} and electrodynamic balance.¹¹ Experiments on dusty plasmas show many-body dynamics of levitating dust particles. An interesting scheme was demonstrated to levitate generic particles above a hot surface by $100\text{ }\mu\text{m}$ based on the Knudsen compressor effect.^{12,13} The authors further demonstrated levitation of ice particles above a Peltier element for minutes.^{14,15}

In this paper, we report an experimental setup to achieve stable, three-dimensional trapping of various particles at pressures of $P = 1\text{--}10\text{ Torr}$. The levitation and trapping do not require external electromagnetic forces or the Knudsen compressor effect and are achieved with only the thermophoretic force in the presence of a strong temperature gradient. We observe stable trapping of various objects including ice particles, ceramic spheres, hollow glass bubbles, polyethylene spheres, and lint. The stability of levitation allows us to measure the levitation heights of the particles as well as the dynamics. Based on a numerical model to describe the temperature distribution in our system, we obtain a good agreement between our experimental observation and theoretical calculation over a large range of temperatures where stable levitation is achieved.

Our experimental setup is displayed in Fig. 1. At the center of the vacuum chamber is a 10-mm gap between a warm copper plate and a stainless steel bucket filled with liquid nitrogen at $T = 77\text{ K}$. The bottom copper plate has a diameter of 33.7 mm and is attached to a Peltier cooler for temperature control. A thermistor is embedded in the plate to allow for temperature monitoring. The stainless steel bucket has a smaller outer diameter of 25.4 mm. The bottom surface of the bucket acts as the top plate and its temperature is monitored by a K-type thermocouple. In the presence of a large temperature gradient, particles are levitated and trapped near the mid-plane between the top and bottom plates.

The levitation comes from the thermophoretic force in our chamber. As illustrated in Fig. 2, air molecules coming

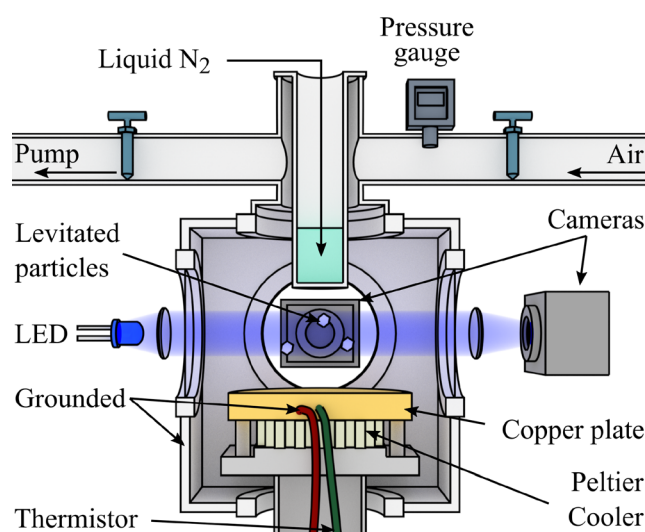


FIG. 1. Experimental apparatus. The copper plate and liquid nitrogen bucket establish a temperature gradient. The pressure is regulated by a vacuum pump and a needle valve that opens to atmosphere. Temperature of the copper plate is measured with a thermistor and regulated with a Peltier cooler. An imaging setup using a blue LED light records the dynamics of the levitating particles. During the levitation, we monitor the temperatures of the top and bottom plates. The plates and the chamber are electrically grounded to avoid electrostatic forces on the particles.

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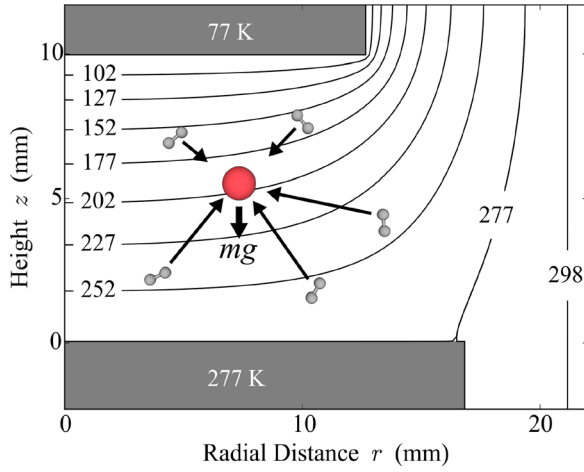


FIG. 2. Temperature distribution in the chamber. Temperature contours are obtained by solving Eq. (2). Boundary conditions hold the top plate at 77 K, bottom plate at 277 K, and chamber wall at 298 K. Collisions with background gas molecules exert a net force opposite to the direction of the temperature gradient owing to the larger momentum transfer from hotter molecules. In our system, particles experience both an upward force counteracting gravity mg and a radial confining force.

from the hotter side transfer on average a larger momentum to the particle than those from the colder side. A net momentum transfer due to collisions with molecules overcomes gravity and pulls the particles toward the symmetry axis of the two plates.

In the presence of a temperature gradient ∇T , the thermophoretic force on a sphere of radius a is given by¹⁶

$$F_{th} = -f_T \frac{a^2 \kappa}{v} \nabla T, \quad (1)$$

where κ is the thermal conductivity of the medium (rarefied air in our case), $v = \sqrt{2k_B T/m}$ is the most probable speed of the air molecules, m is the molecular mass, k_B is the Boltzmann constant, and f_T is a dimensionless thermophoretic parameter that depends on the Knudsen number $Kn = \ell/a$. The Knudsen number is defined as the ratio between the molecular mean free path $\ell = \mu v/P$ and a , where μ is the dynamic viscosity. In the free molecule regime where $Kn \gg 1$, f_T reaches the maximum value of $16\sqrt{\pi}/15 \approx 1.89$. In the hydrodynamic limit where $Kn \ll 1$, however, f_T approaches zero and the thermophoretic force vanishes.¹⁶

The thermophoretic force described in Eq. (1) is associated with heat transfer in the thermal conduction regime. In a temperature field $T(\vec{x})$, heat flows according to Fourier's law as $\vec{Q} = -\kappa \nabla T$. Associated with the heat flow, momentum flow per unit area can be estimated as $\dot{p}/A = \vec{Q}/v$, where A is the effective cross section of a particle. The force on a sphere with radius a is thus $F = \dot{p} = -\pi(a^2 \kappa/v) \nabla T$ and is consistent with Eq. (1) in the molecular flow regime.

To stably levitate and trap a particle, one requires a restoring force toward the trap center for small displacements in all directions. It may appear that a three-dimensional minimum in the temperature field $T(\vec{x})$ is required. Such condition, however, cannot be satisfied without a heat sink under the condition that $\nabla \cdot \vec{Q} = 0$ (Earnshaw's theorem). Together with Fourier's law, the temperature field in free space satisfies

$$\nabla \cdot (\kappa \nabla T) = 0. \quad (2)$$

Since the thermal conductivity of air increases with temperature,¹⁷ the above equation gives $\nabla^2 T = -|\nabla T|^2 \kappa'(T)/\kappa(T) < 0$, which excludes the possibility of stably confining a particle in free space based solely on a temperature gradient.

The requirement for stable thermophoretic levitation is given by $\nabla \cdot (F_{th} + F_G) < 0$, where the F_G is the gravitational force. Using Eqs. (1), (2), and $\nabla \cdot F_G = 0$ in free space, we obtain the condition for stable levitation

$$-\nabla \cdot F_{th} = \kappa a^2 |\nabla T|^2 \frac{d f_T}{d T} \frac{1}{v} > 0. \quad (3)$$

Since thermal conductivity κ is positive, the above condition is equivalent to a sufficiently strong temperature dependence of f_T given by $d(f_T/\sqrt{T})/dT > 0$.

An example of the temperature distribution in our system is shown in Fig. 2 and is numerically calculated based on Eq. (2) using code written in Python specifically for this purpose. Here, one can see a positive curvature of temperature in the radial direction that offers radial confinement. In the vertical direction, however, the curvature of the temperature is negative, shown in Fig. 3(a).

The key to stabilizing particles in the vertical direction is the thermophoretic parameter f_T , which is strongly suppressed when particles enter the hydrodynamic regime. Since our system is isobaric $p = nk_B T = \text{const.}$, lower temperature near the colder top plate suggests higher density. Particles moving near the top plate are thus deeper in the hydrodynamic regime with a corresponding smaller thermophoretic force. The total force thus switches sign when the particles are over-levitated. The distribution of temperature, temperature gradient, and levitation force in our system for a typical operation is shown in Fig. 3.

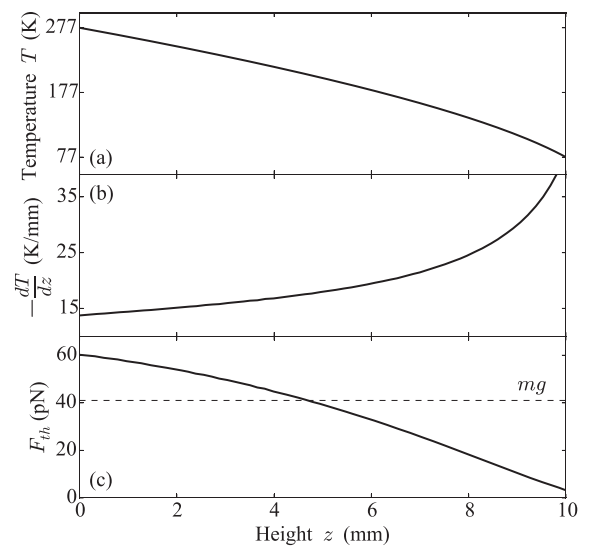


FIG. 3. Thermodynamic quantities and thermophoretic force simulation for a $10 \mu\text{m}$ polyethylene sphere with $\rho = 998 \text{ kg/m}^3$ and $P = 6 \text{ Torr}$. The temperature (a) and temperature gradient (b) are plotted based on the parameters in Fig. 2. (c) Thermophoretic force (solid line), calculated from Eq. (1), is compared with the gravitational force on the particle (dashed line). The intersection of the two curves indicates the trap position of the particle. The weaker levitation force as the particle ascends offers the levitation stability.

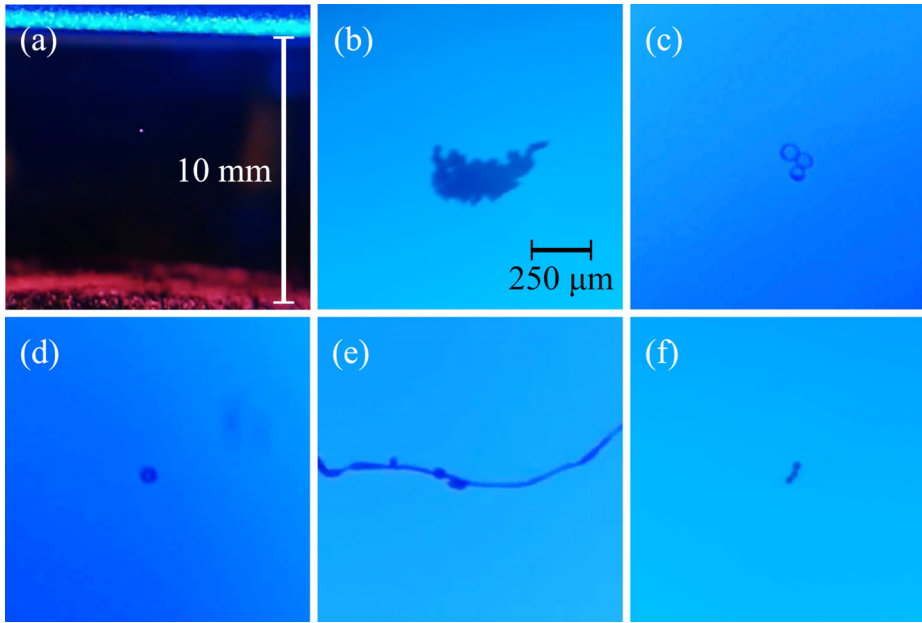


FIG. 4. Gallery of levitating particles. (a) Low magnification image of one polyethylene sphere (Cospheric, UVPMS-BR-0.995). High magnification images of (b) one ice particle, (c) an aggregate of three hollow glass bubbles (Miapoxy, 406354), (d) one ceramic sphere (Miapoxy, 406344), (e) one piece of lint, and (f) an aggregate of three polyethylene spheres.

The experimental procedure goes as follows. We first deposit the desired particles on the bottom copper plate. After sealing the chamber, we pump the chamber down to $P = 1\text{--}10$ Torr. A pressure gauge (Thyracont, VD8) continuously records the pressure throughout the experiment. We then add liquid nitrogen to the steel bucket and the top plate quickly reaches 77 K. We launch the particles by vibrating the chamber, and once above the bottom plate the particles experience the upward thermophoretic force. Depending on the strength of vibration, the pressure, and the type of particles, the number of levitated particles ranges from one to hundreds. Typically ceramic spheres and hollow glass bubbles tend to be levitated in greater numbers than polyethylene spheres. We also observe stably levitating lint as long as 1 mm in length. Figure 4 displays a gallery of levitating objects of different types.

The levitation procedure for ice particles differs from the above. The top plate is cooled down to 77 K before we seal the chamber. Water vapor condensing on the top plate creates a layer of ice particles. The chamber is then pumped down to the desired pressure. Upon external vibration of the chamber, ice particles fall from the top plate and some are levitated.

Our detection system consists of two digital cameras along with a light-emitting diode (LED) that illuminates the levitating particles. One camera with high magnification faces the collimated LED beam and provides a closeup view of the particles. Representative images are shown in Figs. 4(b)–4(f). The second camera records the dynamics of particles with a full field of view including the top and bottom plates, shown in Fig. 4(a).

To quantitatively characterize the levitating force at play, we analyze the measured height of singly levitating polyethylene spheres. We focus on single particle levitation to avoid the complex dynamics of multiple particles. In this case, particles remain stably levitated for our longest observation times of one hour, giving us ample time to study the conditions of levitation. To aid comparison with theoretical calculations,¹⁶ we focus on spherical particles. After examination with a microscope,

polyethylene spheres were chosen as they appeared the closest to ideal spheres.

Since levitating particles are easily affected by air currents resulting from pressure changes, we keep pressure constant and vary the temperature of the top plate. To achieve this, we allow the liquid nitrogen in the steel bucket to evaporate, which slowly warms up the top plate over ~ 10 min. During this time, we synchronously record the temperatures, pressure, and particle height.

Examples of singly levitating polyethylene spheres are shown in Fig. 5, where the top plate temperature slowly increases from 77 K toward room temperature. The particle's levitation height gradually decreases as the top plate warms up because of the decreasing temperature gradient in the chamber.

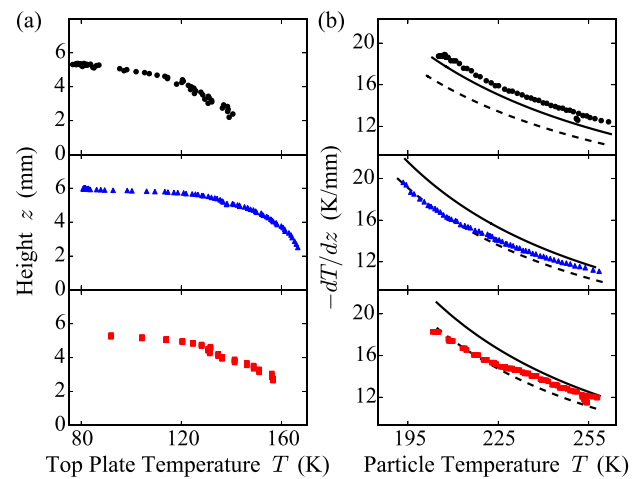


FIG. 5. Levitation of polyethylene spheres during a temperature ramp. (a) After levitation, we allow the top plate to warm up and record the levitation height. Circles (black) correspond to a sphere of radius $a = 12.7\ \mu\text{m}$ at $P = 3.9$ Torr, triangles (blue) correspond to one with $a = 9.8\ \mu\text{m}$ and $P = 6.4$ Torr, and squares (red) correspond to one with $a = 9.3\ \mu\text{m}$ at $P = 7.3$ Torr. (b) Temperatures and temperature gradients at the location of particles. Theoretical calculations are from the studies of Takata (solid line)¹⁸ and Yamamoto (dashed line).¹⁹

Levitation height is given by the balance of gravity and the thermophoretic force $F_{th} = \rho V g$, where $\rho = 998 \text{ kg/m}^3$ is the density of polyethylene spheres, $g = 9.8 \text{ m/s}^2$ is the gravitational acceleration, and V is the particle volume determined from the particle radius. To calculate the thermophoretic force, we determine the thermodynamic quantities at the location of the particle and extract the thermophoretic parameter f_T for a specific Knudsen number Kn from Eq. (1) and $F_{th} = \rho V g$.

To determine the thermodynamic quantities of air at the particle's location, we solve the air temperature distribution in the chamber using the steady state heat diffusion equation, Eq. (2), with the appropriate boundary conditions. From the distribution, we compute the temperature and temperature gradient at the location of the particle based on the approximate functional form, as well as the thermal conductivity κ and dynamic viscosity μ of rarefied air based on the approximate functional forms.¹⁷ For comparison, theoretical expectations of the temperature gradient necessary to cancel the gravitational force are calculated based on the models by Takata *et al.*¹⁸ and Yamamoto and Ishihara.¹⁹ Finally, the Knudsen number $Kn = \ell/a$ is also evaluated at the position of the particle. The procedure is repeated during the entire warming up process. The measurements on three independent polyethylene spheres are shown in Fig. 6.

Our measurements are consistent with the theories of Takata *et al.*¹⁸ and Yamamoto and Ishihara,¹⁹ but not with Talbot *et al.*²⁰ The discrepancies of 5% – 10% between the measurements on the three particles are caused by systematic uncertainties in measurement. The particle size determination suffers from an uncertainty of approximately $1 \mu\text{m}$ due to limited imaging resolution. The two plates do not have perfectly uniform temperature distribution, and the particles can levitate slightly away from the symmetry axis. These result in uncertainties of 5% – 10% in the determination of temperature and temperature gradient.

Another systematic effect is the photophoretic force coming from external radiation.^{21,22} Such force is significant

in the former levitation experiments.^{14,15,22} We estimate the photophoretic force in our system using $I = \epsilon \sigma T^4$, where ϵ is the emissivity and σ is the Stefan-Boltzmann constant, and find that the radiation intensities of 10 W/m^2 from the copper plate at 277 K and 2 W/m^2 from the top plate at 77 K contribute a net levitation force of $<30 \text{ fN}$ on the particles. Compared to the typical particle weights of $25\text{--}100 \text{ pN}$, the photophoretic force is negligible.

Convective flow is also a potential systematic, which occurs due to the Rayleigh-Bénard instability. In a rarified gas, convection occurs at high Rayleigh number^{23,24}

$$Ra = \frac{2048}{75\pi} \frac{1-r}{(1+r)^2} \frac{1}{FrKn^*} > 1708. \quad (4)$$

Here, $Kn^* = \ell/H$ is the Knudsen number of the system, $Fr = v^2/gH$ is the Froude number, $r = T_c/T_h$ is the temperature ratio of the two plates, and H is the plate separation. Given typical parameters of $T_c = 77 \text{ K}$, $T_h = 277 \text{ K}$, $\ell = 7 \mu\text{m}$, $H = 1 \text{ cm}$, and $v = 290 \text{ m/s}$, our Rayleigh number $Ra = 9.1$ is well below the critical value of 1708. Thus, the convection is insignificant.

In summary, we demonstrate an approach to levitate various particles with thermophoretic force. Particle levitation enhances our capability to detect feeble forces on macroscopic objects and distinguishes theoretical thermophoretic force models. When two and more particles are levitated, interesting dynamics in the presence of interparticle interactions and external forces can be seen, offering new possibilities to study many-body systems in a microgravity environment.

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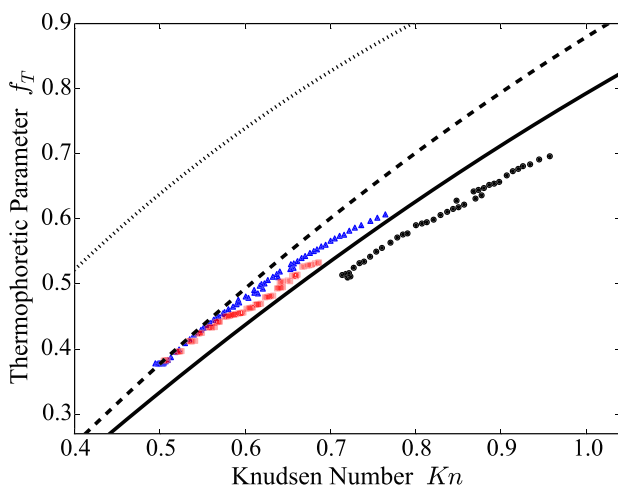


FIG. 6. Comparison between f_T from experimental measurements and theoretical models. Thermophoretic parameter f_T calculated from the data in Fig. 5 is compared with theoretical calculation by Takata *et al.*¹⁸ (solid line), Talbot *et al.*²⁰ (dotted line), and Yamamoto and Ishihara¹⁹ with $k_{12} = 0.03$ (dashed line), where k_{12} is the ratio of the thermal conductivity of gas and that of the levitating particle.

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